

Imitative (herd) behaviour in direction decision-making hinders efficiency of crowd evacuation processes



Milad Haghani^{a,b,*}, Majid Sarvi^b

^a Institute of Transport and Logistics Studies, The University of Sydney Business School, Australia

^b Department of Infrastructure Engineering, The University of Melbourne, Australia

ARTICLE INFO

Keywords:

Agent-based models
Behavioural contagion
Indoor evacuation modelling
Simulated escape behaviour
Social influence
Peer imitation

ABSTRACT

Humans use social information to improve their decisions in scenarios that involve uncertainty. Previous research has shown that this social influence can have positive or negative outcomes depending on the context of decision and characteristics of the system. A particular scenario where the social information matters is the collective escape where behaviour of others could be used as an extra source of information and thus, influence the decision-making. The question is whether there could be any benefit for escaping human crowds to display imitative tendencies (or the so-called 'herd' behaviour) in their decision making. Here, we investigate this question in relation to exit direction decision-making through extensive computer simulation analyses. Using parametric models of decision making, we increased the imitative tendency of (behaviourally) homogenous crowds on a continuum, simulated the escape process and outputted measures of system efficiency. We examined a broad range of simulated scenarios in terms of the shape of the environment, exit visibility and spatial distribution of exit capacities. Results showed that for none of the scenarios we tested, the escaping crowd could benefit from amplified tendencies to imitate peers. Our numerical analyses suggested that, at least when familiar with escape routes, there is no rationale for blindly following the direction choices of the crowd in emergency scenarios. Strong tendency to copy direction choices of the crowd could substantially hinder the process of evacuations and increase the risk of casualty. These findings have potential implications for evacuation training guidelines. They provide insights into how individuals should be guided/instructed to conduct themselves in case of an emergency in order to facilitate and optimise the evacuation process. A question for future research is how imitation (herding) in other aspects of escape behaviour (such as movement initiation or adapting (changing) direction choices) can influence the system.

1. Introduction

1.1. Herding as a broader concept

Making efficient strategic decisions is a crucial element for the survival (or thrive) of many species including humans. In making such decisions, and in particular when the decision context involves certain degrees of uncertainty, humans tend to utilise all possible sources of information notably the social information. Humans often resort to the decisions made by others as extra sources of information to improve their decision making. When this process is followed by a tendency to 'imitate' the decision of the majority, it could lead to an unplanned coordination of the actions which is often referred to as 'herd' behaviour (Raafat et al., 2009; Zhao et al., 2011). The term and the underlying phenomenon is not merely limited to the cases of decision

making. Rather in broader terms, this effect might happen in many aspects of social interactions like the alignment of thoughts or beliefs (Farrell, 2011; Lorenz et al., 2011; Moussaïd et al., 2013; Wang et al., 2014).

From a linguistic perspective, the term 'herding' does not necessarily equate 'imitation'. However, in many contexts in the scientific literature, these two terms have been to great degrees intertwined. The definition provided by The Longman Dictionary of Contemporary English acknowledges the closeness of these two concepts for instance by equating the 'herd instinct' to "*the need to behave in the same way as everyone else does*" (Mayor, 2009). The herding phenomenon is a relatively well-documented aspect of the human behaviour in economic and finance domains (Andersson et al., 2009; Baddeley, 2010; Huang and Chen, 2006; Kaltwasser, 2010; Kim et al., 2004; Kukacka and Barunik, 2013; Manahov and Hudson, 2013; Prechter, 2001; Yook and

* Corresponding author at: Institute of Transport and Logistics Studies, The University of Sydney Business School, NSW 2006, Australia.

E-mail address: milad.haghani@sydney.edu.au (M. Haghani).

<https://doi.org/10.1016/j.ssci.2018.12.026>

Received 11 June 2018; Received in revised form 29 October 2018; Accepted 21 December 2018

Available online 07 February 2019

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Kim, 2008). According to major influential studies in economic literature, the term ‘herd behaviour’ has been frequently used with a connotation of ‘imitative behaviour’. For example, the seminal study of Banerjee (1992) refers to the term ‘herd behaviour’ as a case in which “people will be doing what others are doing rather than using their information”. In the domain of investment decision making, Scharfstein and Stein (1990) use the term herd behaviour as circumstances in which “managers simply mimic the investment decisions of other managers, ignoring substantive private information”. This behaviour has been identified as a major cause of turmoil and volatility in financial markets (Wagner, 2003). Previous studies in modern economic literature have identified sociological drive that encourages investors (especially in uncertain situations) to believe and to do what others do (Baddeley, 2010). The appreciation of such effects on financial behaviour of humans has been a major factor in recent shifts from classic economic models towards models of bounded rationality (De Martino et al., 2006; Grüne-Yanoff, 2007; Oshry et al., 2006; Simon, 1986).

The herding phenomenon is not limited to the financial contexts and has been recognised as a major influential factor that drives humans behaviour in contexts like fashion (Bikhchandani et al., 1992), political orientation (Battaglini, 2005), personal beliefs (Teraji, 2003), mass hysteria (Balaratnasingam and Janca, 2006) or decision making in collective evacuations (Carlson et al., 2014). The behaviour has been widely observed in non-human animals as well, especially in the context of predator avoidance or food search (Couzin et al., 2005; Hamilton, 1971).

1.2. Herding in direction choices

One particular area of decision making in which both humans and animals are heavily exposed to the decisions of other individuals in their groups is the choice of direction in mass movements (Albi et al., 2016; Arganda et al., 2012; Faria et al., 2010; Nasir et al., 2014). An example of mass movements could be escaping from a predator or a life-threatening threat (Koo et al., 2014, 2013; Manley and Kim, 2012; Wagner and Agrawal, 2014). As shown by Arganda et al. (2012), when animals move in groups, the behaviour of other animals will become extra sources of information and this information is incorporated in their decisions in different ways. They investigate different rules that various species take into account in their own choice of direction the number of animals taking each directional option (Arganda et al., 2012).

The role and the significant influence of the social interactions on the emergent escape pattern is undeniable (Albi et al., 2016; Lovreglio et al., 2016b). However, the direction of the influence has so far been a rather mysterious topic (Wang et al., 2015) with the mainstream research suggesting the idea that humans would act in an imitative way in such scenarios similar to animal groups (Chu et al., 2014; Pan et al., 2006). The idea originated predominantly from models of ants or mice experimentations as a proxy for human crowds’ escape behaviour (Altshuler et al., 2005; Saloma et al., 2003).

1.3. Herding in evacuations

A particular area of interest in studying the herd behaviour among all domains to which this phenomenon pertains is cases of emergency evacuations (Bernardini et al., 2017, 2016; Haghani and Sarvi, 2017c; Ronchi and Nilsson, 2013; Ronchi et al., 2014; Shahhoseini et al., 2018, 2017). In addition to the term ‘herding’ or alternatively ‘herd-like’ or ‘herd-type’ behaviour (Bode and Codling, 2013; Fang et al., 2016; Haghani et al., 2017; Lovreglio et al., 2016b, 2014b; Van den Berg et al., 2018) which was originally used in this context by Helbing et al. (2000) in describing a “tendency towards mass behavior, that is, to do what other people do” (page 487), authors have used a variety of language and terminologies to describe such phenomenon. This includes ‘symmetry breaking’ (Altshuler et al., 2005; Chung and Lin, 2017; Li

et al., 2014), ‘peer imitation’ (Moussaïd et al., 2016), ‘social influence’ (Kinader et al., 2014a, 2014b), ‘peer influence’ (Haghani and Sarvi, 2017b), (copying) ‘neighbour behaviour’ (Kinader et al., 2018) and ‘follow-the-crowd behaviour’ (Haghani and Sarvi, 2017a).

1.4. Herding, beneficial or detrimental?

Whether or not herding has a beneficial or detrimental effect for the group (Zhao et al., 2011) has been a traditional question for many systems. The dominant view in the literature is that herding has negative effects on the system. It can ruin the balance of equilibriums in complex adaptive systems and cause excessive volatility, bubbles or crashes in financial markets and thus financial crises (Wagner, 2003). Other studies have also shown how social influence can narrow down the diversity of opinions to the extent of undermining the wisdom of the crowd (Lorenz et al., 2011).

The existing evidence about the effect of herding on complex systems, however, is not unanimous (Farrell, 2011). The study of Zhao et al. (2011) particularly challenged the belief that formation of a herd necessarily ruins the balance of resource allocation and thus cause detrimental effects. Their study suggested that the effect does in fact depend on the ratio of the resources. When there is bias in resource ratios, the formation of a moderately sized herd can benefit the system, according to this study.

The knowledge about this particular question (i.e. potential benefits of herding) is even further sparser and highly speculative in relation with humans escape behaviour (Gwynne and Hunt, 2018). While many simulation models of humans escape have attempted to formulate and incorporate herding behaviour in their methodologies (Chu et al., 2014; Helbing et al., 2000; Pan et al., 2006, 2007) (a practice which, as we will show in this work, can potentially generate highly unrealistic outcomes), no study has systematically investigated potential benefits or detriments caused by this phenomenon. The question is, would there be circumstances under which escaping human crowds may benefit from the herd behaviour?

In this section, we aim to provide an overview of the common beliefs about the (in)effectiveness of herding in escape scenarios just by quoting a number of statements from a range of influential articles in this area. It has been suggested by Bohannon (2005) that “the first thing we do in an emergency situation is look to each other for support and information. But that response slows movement dramatically. On a larger scale, people form groups similar to animal herds in which individuals let the crowd do the navigating, often passing right by exits within clear view” (page 221). In another study, Pan et al. (2007) stated that “one well-known example of social proof under emergency situations is the herding behavior—when under highly uncertain and stressful situations, an individual tends to follow others almost blindly. Sometimes herding behavior helps people to exit safely, and at other times, the herding behavior may lead people to a dead end or cause the blockages of some exits even though other exits are not fully utilized. This is a particularly interesting phenomenon in crowd dynamics and the phenomenon has now been incorporated in some computational models” (page 119). Pan et al. (2006) claim that “building designers often assume that a crowd would exit evenly among multiple exits of a room in case of an emergency; herding behavior contradicts such an assumption” (page 459). In one of the most recent studies that have addressed this topic, Moussaïd et al. (2016) have pointed out that “it remains unclear to what extent pushing, overcrowding and peer imitation can affect the efficiency of egress. The main obstacle to answering these questions is the scarcity of detailed empirical data” (page 2). This is the primary question that our study is driven by. In addition, we also address the common view in the literature that states by not incorporating the herding behaviour in the modelling framework, simulation models underestimate the escape time of human crowds. We provide evidence to the contrary of this assumption in this work, showing that models that produce herd flow in direction decisions grossly exaggerate the escape time.

1.5. The current study

In this work, we investigate the potential benefits of herd behaviour in escape scenarios using computer simulations. We exclusively focus on exit directional choice making (Casareale et al., 2017; Fridolf et al., 2013; Haghani et al., 2018a; Lovreglio et al., 2014a, 2016b) and consider evacuations of confined crowded spaces. We examine various scenarios under the symmetric and asymmetric distributions of exit capacities to identify whether an escaping crowd could potentially benefit from following the majority in certain cases (an idea speculated by previous studies as discussed earlier). Using a modelling structure formulated in our earlier study (Haghani and Sarvi, 2017a), we were able to gradually (almost on a continuum) alter the imitative tendency of the individuals, while simulating the system and measuring the system performance in terms of conventional metrics of efficiency. As a secondary analyses, in this study, we use empirical observations from a limited held-out sample of a more recent experiment (Haghani and Sarvi, 2018c) to test the effect of model misspecification (in terms of imitative tendencies) on prediction outcomes. We demonstrate the extent to which the predictions of a model based on the herding rule can deviate from the reality.

In the following section, we detail the simulated scenarios that we set as the benchmark for our numerical tests in this work. We test twenty different scenarios that cover a broad variety of geometric features to ensure that the resultant outputs are not specific to the geometry of the simulated scenario. We then describe the simulation tool that we employed for performing the numerical tests (as a multi-layer simulation model of evacuation) while focusing primarily on the exit direction choice module. We detail how such parametric model allows us to increase the herding tendency on a continuum and measure the simulated effect at an aggregate level. Section 4, the analysis section, reports the outputs of these numerical tests. As a secondary and minor part of this study, we also examine the simulation model under various degrees of imitative tendency to replicate (at aggregate level) a limited number of experimental observations. The purpose of this section is to demonstrate how mis-specification of the tendency to imitate neighbour behaviour may result in misleading simulation outputs. Section 5 summarises the conclusions and section 6 provides detailed discussions as well as directions for future work.

2. Methods

2.1. Simulated setups

In our computer simulated scenario of escape, we examine twenty different setups in order to test the potential benefits or detriments of herd behaviour in escape situations. These setups differ from one another based on the general geometry of the simulated space, the exit capacities and their distribution in that environment, the presence of barricades and the visibility of exits. In total, we examined five different types of geometry and for each given geometry type, we examined four different spatial distribution patterns of exit capacities (or exit widths). Figs. 1 and 2 show these general geometries in the form of still images taken from the process of simulation while calculations were in progress. Each geometry consists of a main room with three exits (the number of exits is constant across the scenarios) and simulates the escape of 400 agents. These agents are generated at a rate of 15 agents per second in random positions in a rectangular auxiliary room that is connected via a wide exit to the main room. In the pictures, these auxiliary rooms are coloured yellow. As soon as being generated in the auxiliary room, agents start to move towards the main room and make directional exit decisions to escape the main room. The process of simulation for each agent starts at the moment the agent is generated in the auxiliary room and terminates once it reaches one of the three final gates in the main room.

Setups 1 to 8 are geometrically symmetric with the second block of

four setups (i.e. setups 5 to 8) differing with the first one by including a barricade in the room that is assumed to create invisibility effect for one of the exits. Setups 13 to 16 are also semi-symmetric with the difference that the inflow of the simulated pedestrians are generated from two auxiliary rooms (rather than one) located at the both ends of the main room. Clearly, 200 agents are generated at each side in order for the scenarios to be comparable with those with only one auxiliary room. There are four barricades in this setup that create the invisibility effects for the exits depending on the side from which the pedestrian enters the main room. Setups 9 to 12 as well as setups 17 to 20 are geometrically asymmetric. For setups 9 to 12, the invisibility effect is created through both barricades and the wall positions in the environment. In setups 17 to 20, there is no barricade and the invisibility effect is exclusively due to the architectural design (the relative position of the walls and exits) with one exit being invisible and the two others being visible. For each block of four setups that share the same geometric features, we created four (sub) setups that differ from one another based on the capacities (or widths) of their exits. These four setups have respectively 0, 1, 2, and 3 exits with 50 cm width while the other remaining exits (out of the total 3 exits) were 100 cm wide. These combinations created setups with spatially balanced as well as setups with spatially imbalanced distribution of exits. They also created scenarios that were moderately crowded (setups with three 100 cm-wide exits) as well as heavily crowded scenarios (where all exits were all 50 cm wide). Table 1 summarises these geometric features for all twenty scenarios that were described above.

2.2. Simulation method

Once a simulated agent steps into the main room (where it needs to decide on its direction of movement), the first decision to be simulated is the choice of exit for that agent. This decision is generated using the model that we formulated and reported in Haghani and Sarvi (2017a), that has now in this work been integrated with a broader structure of modelling, enabling us to generate movements and thus simulate scenarios. The formulation of the model is presented in Eq. (1). The model in total has five variables that along with their coefficients determine the utility of each exit direction. Of these five variables, three represent the social influence (the effect of other escapees that are either moving towards exits or queuing at exits) and two are related to the physical factors (spatial distances to exits and exit visibility). The (magnitude and sign of the) coefficients of these variables in the utility function are the reflection their corresponding variable's weight on agent's perception of the desirability of each alternative direction. In other words, they quantify the relative valuation of each attribute for exit directions (also known as the marginal utility of that attribute). The choice probabilities are generated using the logit function formula (Eq. (2)).

$$\begin{aligned}
 V_{exit\ i} = & \beta_1 (\text{distance to exit } i) + \beta_2 (\text{size of crowding near exit } i) \\
 & + \beta_3 ((\text{size of flow moving to exit } i) \times (\text{visibility of exit } i)) \\
 & \quad \quad \quad = (\text{size of flow moving to visible exit } i) \\
 & + \beta_{4n} ((\text{size of flow moving to exit } i) \times (1 - (\text{visibility of exit } i))) \\
 & \quad \quad \quad = (\text{size of flow moving to invisible exit } i) \\
 & + \beta (\text{visibility of exit } i)
 \end{aligned} \tag{1}$$

$$Prob(exit_i) = \frac{\exp(V_{exit_i})}{\exp(V_{exit_i}) + \exp(V_{exit_j}) + \exp(V_{exit_k})} \tag{2}$$

For the purpose of the simulation analyses and to create various degrees of herding (imitative) tendencies, we altered the coefficients of the three variables related to social influence jointly and incrementally with small step sizes. However, the base coefficient values that we departed from were the parameters calibrated from the previous experimental data reported in Haghani and Sarvi (2017a). In doing that, we increased and decrease the herding tendency (i.e. the valuation of the social factors, reflected collectively in β_2 , β_3 and β_4 parameters in

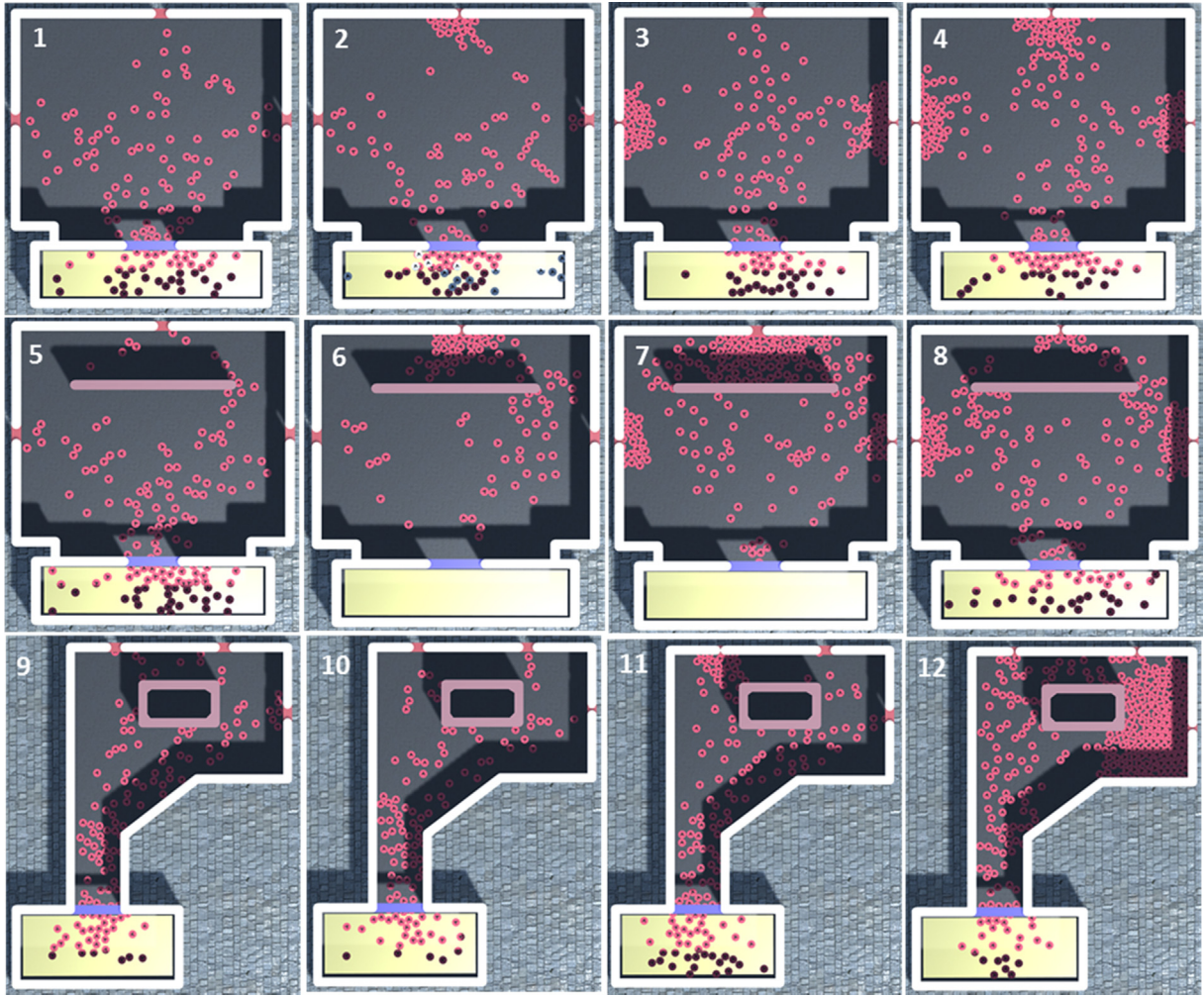


Fig. 1. The simulated escape setups for setups 1 to 12. Each image shows a snapshot from the visualisation of the process of the simulation for the setup that it represents. Simulated agents are generated at a fixed rate in the auxiliary room (shown in yellow rectangle in each image) connected to the main room via a wide exit. Each room has three main exits whose widths vary across different setups. Each scenario simulates the escape of 400 simulated agents. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Eq. (1)) through an α factor (while keeping the value of the two other parameters constant). Assuming that β^* is the value of the estimated parameter (for any of the three aforementioned parameters), in our simulation, we visited all parameter values generated by the function $\beta_{1,2,3} = \beta_{1,2,3}^* \pm (\frac{\alpha}{100})|\beta_{1,2,3}^*|$, while α ranges from -500 to $+500$. In this formulation, $\alpha = 0$ would be corresponding to the ‘natural’ behaviour (or tendency) that we observed and estimated in previous experiments,

$\alpha = +500$ would be reflective of an extreme tendency for herding (i.e. imitating the exit chosen by the majority of the simulated agents at the moment that the simulated choice maker chooses its exit); and $\alpha = -500$ would be associated with an extreme degree of avoid-the-crowd behaviour (i.e. opposite of imitative behaviour) in which each simulated agent chooses (at the moment of their decision) the exit that is chosen by the least number of agents at that moment. To cover the

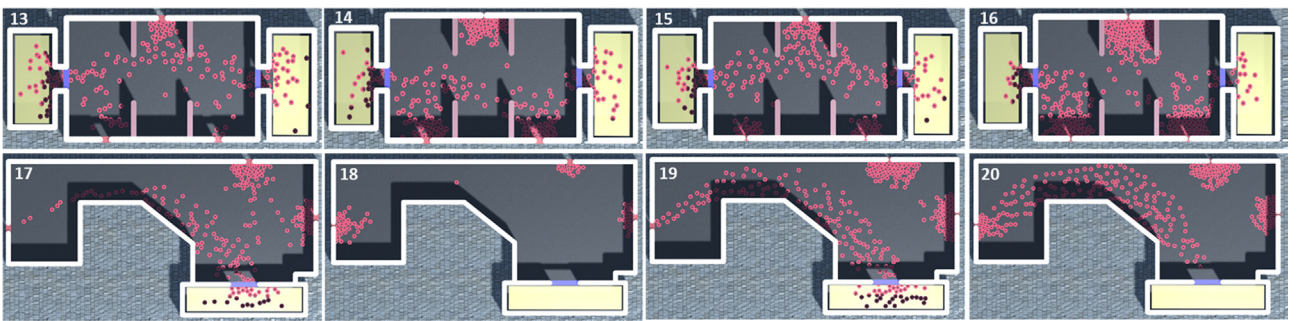


Fig. 2. The simulated escape setups for setups 13 to 20. Each image shows a snapshot from the visualisation of the process of the simulation for the setup that it represents. Simulated agents are generated at a fixed rate in the auxiliary room(s) (shown in yellow rectangle(s) in each image) connected to the main room via a wide exit. Each room has three main exits whose widths vary across different setups. Each scenario simulates the escape of 400 simulated agents. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1

Details of the simulated scenarios. Each scenario simulated 400 agents escaping through three alternative exits. Each of these exits is either 100 cm or 50 cm wide.

Scenario No.	Geometry type	No. of barricades	No. of invisible exits	No. of narrow exits (w = 50 cm)	No. of entrances
1	Symmetric	0	0	0	1
2	Symmetric	0	0	1	1
3	Symmetric	0	0	2	1
4	Symmetric	0	0	3	1
5	Symmetric	1	1	0	1
6	Symmetric	1	1	1	1
7	Symmetric	1	1	2	1
8	Symmetric	1	1	3	1
9	Asymmetric	1	2	0	1
10	Asymmetric	1	2	1	1
11	Asymmetric	1	2	2	1
12	Asymmetric	1	2	3	1
13	Symmetric	4	2	0	2
14	Symmetric	4	2	1	2
15	Symmetric	4	2	2	2
16	Symmetric	4	2	3	2
17	Asymmetric	0	1	0	1
18	Asymmetric	0	1	1	1
19	Asymmetric	0	1	2	1
20	Asymmetric	0	1	3	1

interval between -500 to $+500$, we used varying step sizes, with the step sizes being smaller for smaller values of α (i.e. when closer to the natural behaviour) and growing as the value of α itself increased (i.e. as we departed further from the natural estimated values of the parameters). In total, the interval was covered by 111 values of α . Given that we had 20 scenarios and that for each value of α in each scenario, we repeated the simulation for 50 times, the entire analyses entailed $20 \times 111 \times 50 = 111,000$ simulation repetitions. For an average of 25 s as the average computation time for each single simulation scenario (based on our computational capacities), this translates into more than 770 computational hours. The work of computation was distributed over 10 standard computers in order to bring the computation time to a reasonable amount. We had 20 different scenarios, therefore, on each computer two scenarios were handled. For any given setup, the program performed 50 repetitions for any given value of α , and once this cycle is finished (and the statistics are outputted), it transitions to the subsequent value of α and performs another cycle of 50 repetitions. The process continues until all 111 values of α are exhausted.

For generating the movements of the simulated agents, a combination of a shortest weighted path algorithm (for pathfinding) (Cui and Shi, 2011) and social force model (for local momentary movement generation (Helbing et al., 2000)) was employed. In this multilayer integrated platform of modelling, once the choice of exit is generated (using the probabilistic method described earlier), the choice is used as an input to generate a path between the position of the agent and the chosen exit. For generating this path, we discretised the space of movement into a system of spatial mesh (see Fig. 3 for an illustration). Each point on this mesh system as well as its immediate adjacent points (by a discounted factor for the adjacent points) is penalised as the result of the presence of barricades (with an extreme amount of penalty) or the presence of pedestrians. See Fig. 3, for a visualisation of the penalty weights overlaid on the grid system for one of our setups. Taking these penalties into account, we generate the shortest path for each agent on this mesh system that connects its current location to the location of the chosen exit and then truncate this path to generate a smooth path and prevent unrealistic movement fluctuations. This path is updated at a certain frequency for each agent to respond to the changes in the dynamics of the environment (the time-dependant factors, like congestion on various parts of the mesh system). See Fig. 3 as a still image that visualises these paths (as well as their truncated version) for all agents

in the scene at one particular moment in time. The information obtained from this level of calculation is then fed to a social force model of pedestrians (Helbing et al., 2000) to generate the actual next step at each time increment of the simulation. The vector connecting the current position of the pedestrian to the next point on the mesh system is inputted as the vector of “desired direction”. This is used to calculate the desired force along with the social and wall forces, the net outcome of which determines the next position of the pedestrian. Fig. 3 also visualises the mass trajectories for the actual movements of the pedestrians in one scenario once the calculations have finished.

As mentioned earlier, for a given setup and a given set of parameters, we ran 50 simulation repetitions. For each run, we calculated and outputted the total evacuation time (the time taken from the onset of the simulation until the moment the last simulated agent reached a final exit) as well as the average of the (simulated) individuals’ evacuation times in that scenario. These quantities were subsequently averaged over each cycle of 50 repetitions and were reported as the outcomes of the analyses.

3. Analyses

3.1. Simulating system efficiency

The outcomes of the simulation analyses have been summarised and plotted in Fig. 4, where the average of total and individual simulated evacuation times associated with each value of α has been plotted for each scenario. The numbers that label each graph are corresponding to the number of the simulation setup that the graph represents. The black continuous lines correspond with the measurements of the total evacuation times and the red dashed lines are related to the average of individual evacuation times. The error bands represent the standard deviations of the measurements. Graphs in each row are related to the setups that share the same type of geometry. Graphs positioned in the same column in this figure are associated with different geometries but are comparable with each other in that they all represent a same pattern of exit capacity spatial distribution. The first and fourth columns are related to the setups that have equal capacities at all three exits. The first column is related to moderately congested scenarios, whereas the fourth column is related to the heavily crowded scenarios (i.e. the highest level of crowding that we examine for each setup by restricting all three exit widths to 50 cm). The two columns in between represent scenarios with imbalanced spatial distribution of exit capacities across the exit points (where a subset of exits is 50 cm-wide and the rest is 100 cm).

There was a virtual consensus in the outcomes of the simulation analyses obtained from the various setups that we examined. According to the line charts in Fig. 4, increasing the tendency to herd almost invariably decreased the evacuation efficiency by prolonging the evacuation process.

With a few exceptions (setup 11, most noticeably), both measurements of the escape efficiency were always consistent at all levels of the herding tendency meaning that both curves showed the same pattern of variation as α changed. This embodies a secondary finding in its own right suggesting that for quantifying the efficiency of the evacuation processes (an essential element of evacuation optimisation (Abdelghany et al., 2014)) either of these two measures may be usable interchangeably and rather indifferently (e.g. as an objective function for optimisation).

According to the majority of the graphs in Fig. 4, we observe the greatest degree for sensitivity of the escape efficiency to the herding tendency in the ranges that are closer to the estimated (or natural) level (observed under experimental conditions). The value $\alpha = 0$ represents the estimated natural tendency (in terms of the marginal valuation of the social-influence variables). As we depart far enough from the natural level of tendency, most curves flatten out indicating a pattern of insensitivity to any more amplification in the follow (or avoid) the

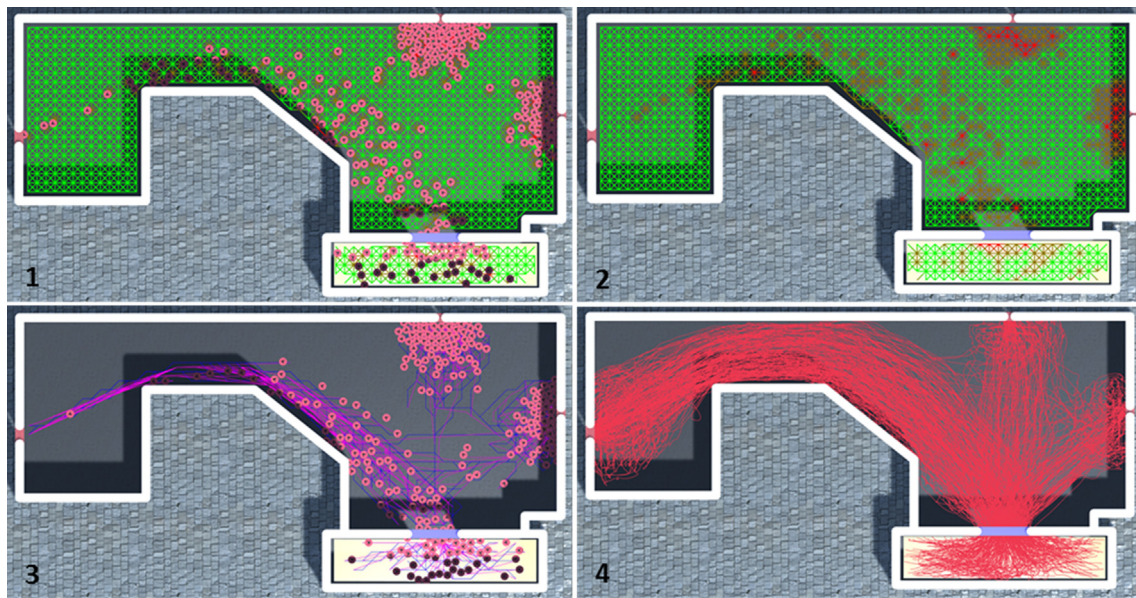


Fig. 3. The process of movement generation for simulated agents once their directional choice is simulated using the logit model. Image (1) illustrates the spatial mesh on which the paths are generated. Image (2) visualises the concept of path penalisation in order to calculate the shortest paths while considering the presence of other agents. Image (3) visualises the shortest paths (and their truncated versions) generated for all agents in the scene between their current positions and their chosen exits. Image (4) visualises the mass trajectories of the simulated movements for all 400 simulated agents once the calculation is over.

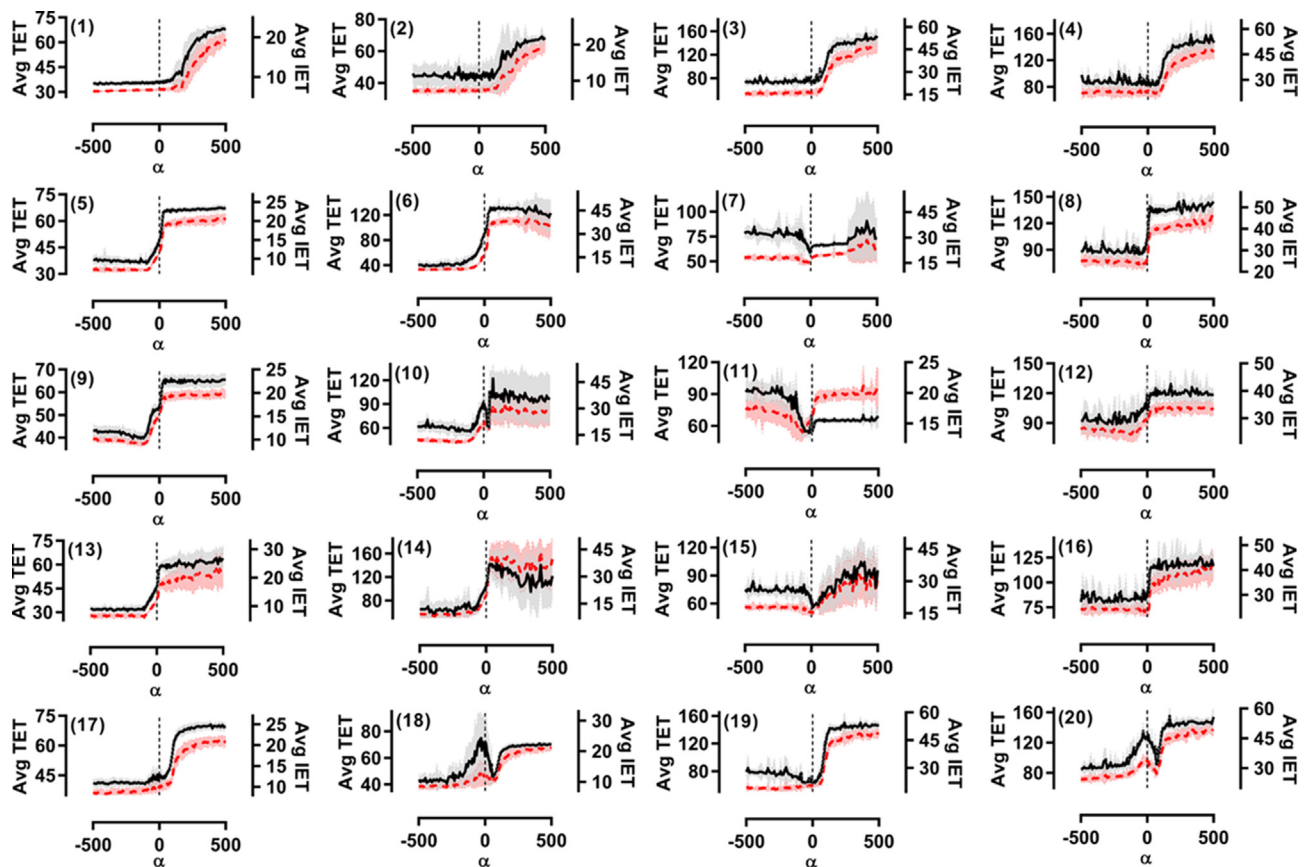


Fig. 4. The outcomes of the simulation analyses for each setup, the variation of evacuation times as the herding tendency changes. The number on each plot is consistent with the number of the simulation setup that the plot represents (given by Figs. 1 and 2). Black continuous and dashed red lines respectively show the variation of the total and average evacuation times (respectively shown on the left and right vertical axes) for various values of the herding tendency (α). The measurements associated for each α value have been averaged over 50 repetitions of simulations. The error bands represent the standard deviations of the measurements. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

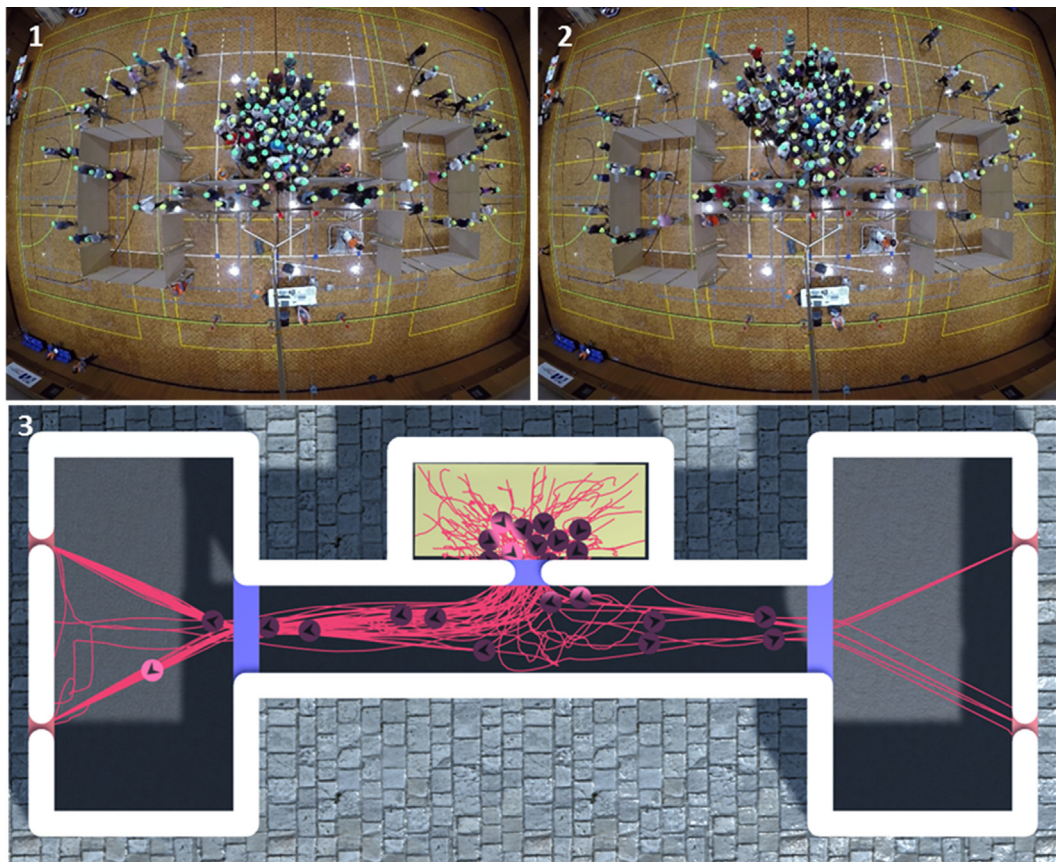


Fig. 5. The experimental and simulated scenarios. Images (1) and (2) are snapshots from the video footage of the experiments in two scenarios. Image (3) is a still image from the process of calculations when those scenarios replicated using the simulation model.

crowd tendency. But in the ranges that are closer to $\alpha = 0$, increasing or decreasing the inclination of agents to herd influences evacuation times at much greater gradients.

While there were exceptions to this observation (especially for setups with imbalanced capacity distributions), the most commonly observed pattern of variation as the valuation of the social factor varies from extremely negative to extremely positive values is as follows. The graph is initially flat which is followed by a sharp increase and then it flattens out again. For certain setups, like 18, 19 and 20 there was certain areas of non-monotonicity, but the dominant and overall pattern was observed to be a flat line, followed by an increase and a relatively flat line again. This indicates that an increase or decrease in the natural tendency to imitate direction choices of others even at small and marginal degrees could substantially influence the outcome and efficiency of the evacuation. Or inversely, one could state that adopting a follow the minority strategy (or an avoid-the-majority) strategy can make significant improvements in the efficiency of evacuation processes almost invariably (considering the case of a homogenous crowd). According to our graphs, the difference between the aggregate outcome of an evacuation process under these two strategies can amount up to nearly 100%. A crowd escape scenario can become twice as much more efficient if individuals in the crowd generally avoid an imitative strategy in their exit direction choices.

Also, a question arises as to whether the level of herding (or no-herding) tendencies that we have estimated from experimental observations (and assume as the natural tendency to herd or not herd) is at the most efficient (or optimum) level. According to the simulation outcomes, for certain setups (like setups 1, 2, 3, 4, 17 or 19), the natural level of marginal utility estimated for social factors is in fact in the range of most efficient (or optimum) level. In those scenarios, the system could not benefit further alternative marginal valuations of the

social influence. According to a number of other setups, however, the crowd could potentially benefit more if they further increased the weights of avoid-the-crowd tendency in their exit decisions. Examples of this were setups 5, 6, 9, 10, 12, 13, 14. In our simulated experiments, we barely observed that the (estimated or) natural humans' valuation of the social factors lies in the least efficient domain.

3.2. Comparison with experimental data

As a secondary analysis in this study, we addressed another question that has implications for modelling of crowd escape behaviour. As the recent literature has suggested through multiple experimental data, the assumption of humans showing noticeable imitative tendencies does not hold under empirical examination. However, since the assumption is entrenched has been accepted as a given fact in the literature, many computational models have tried to implement it in their model formulation and implementations (Chu et al., 2014; Pan et al., 2007; Shiwakoti et al., 2011; Wang et al., 2016). Here, we look at this topic from the perspective of the extent of the (potential) bias that mis-specification of the imitative tendency (which we think of as a mis-characterisation of the behaviour) may cause in the predictions.

In quantifying this bias, we resorted to a limited held-out set of experimental data from a more comprehensive set of empirical tests that we performed to investigate herding behaviour assumption in direction choice making (Haghani and Sarvi, 2018c). The experiment was approved by the Engineering Human Ethics Advisory Group of the University of Melbourne. The experimental scenarios that we employed for this analysis were performed beyond the scope of the main experimental design reported in (Haghani and Sarvi, 2018c). They are regarded 'extra scenarios' that were performed in addition to the main design and to be analysed in a limited way (only at aggregate level) for

model verification purposes such as those reported in this work. Here, we employ these extra scenarios just to serve as a basis (and an extreme example) for quantifying the extent of bias when one mis-specifies the imitative tendency in a simulation model. We merely attend to the macro-scale measurements (i.e. total evacuation times) from this small set of experimental observations. Details of the experiments are accessible in the other study linked to the topic of this article (Haghani and Sarvi, 2018c).

For the sake of this analysis, these empirical tests can be seen as four experimental scenarios in which 117 participants had to wait at an entrance door, entre a built environment (in shape of a maze) and escape it assuming that they escape from a danger. The built environment includes a corridor that connects to two rooms (one room at each end) and the rooms at each end have either one or two exits (see Fig. 5 for still images from the footage of the experiments that also show this geometric setup). In each scenario, either both rooms have two symmetrically located exits, or one room has one of its exits closed. Exits are each 50 cm wide and the entrance is 150 cm wide. We performed four of such (additional/extra) scenarios during that experiment. We had two treatment types, one in which we started the escape process by playing a loud siren (that continued during the entire scenario) and asking participants to imagine they are escaping from an imminent life-threatening danger. The other treatment was related to scenarios in which no siren was played and participants were asked to assume an evacuation but not under a life-threatening danger. For these additional scenarios, participants were simply asked to use their imagination to differentiate between the two contexts (whereas, for the main body of the experiment, monetary incentive was on offer during the stressful treatments) (Haghani and Sarvi, 2018c). We performed each of these treatments twice, once with 4 available exits and once with three available exits. Extra scenarios 1 and 2 had three available exits (under imminent and non-imminent hypothetical danger, respectively) and extra scenarios 3 and 4 had four exits (again, under the assumption of imminent and non-imminent danger, respectively). Table 2 summarises the details of these four scenarios.

Participants did not know about the availability of the exits before performing the scenarios. The setup had been designed in a symmetric way to isolate (to the most possible extent), the effect of the social influence from the physical factors on direction choices. In other words, the idea was to create setups in which the physical attributes are the same for each directional alternative so these factors do not entre in a heavy way in a trade-off with the social factors and so we can assume that the direction choices are predominantly determined by the social influence. Each participant had to make two choices in order to escape that environment, first a directional choice of corridor and second, the choice of an exit (unless they had chosen a corridor that led them to a room with only one exit in that scenario).

We used the observed total evacuation times of those scenarios and contrasted them with the simulated evacuation times under varying degrees of imitative tendency (see Fig. 5 for still an image of the simulation of these scenarios). We used the same directional choice model described by Eq. (1) and Eq. (2), except, here we set all coefficients arbitrarily. Firstly, we switched off the effect of physical factors by setting their coefficients to zero (as we assumed that their effect is negligible on the simulation of these scenarios due to the physical symmetry at each direction decision point). And secondly, we used

equal coefficients for all three social variables. In other words, the model simply measures the number of agents that are already in the corridor (for the corridor choice) when the choice maker steps in the corridor; or the number of agents already in the room (for the exit choice) when the choice-maker agent steps in one of the rooms. The model then weights their influence through a simple coefficient. We changed the value of this single coefficient (i.e. the coefficient for the social influence as described above) from $\beta = -0.5$ (associated with an avoid-the-crowd tendency) to $+0.5$ (associated with a strong herding tendency) at increment sizes of 0.1.

We measured the total escape time for each of the four experimental scenarios explained above. The values have been reported in Table 2. We replicated those scenarios using the simulation method described earlier and measured the simulated total egress times (for 117 simulated agents). We repeated each of the scenarios at each parameter value for 100 times (requiring 4400 simulation runs in total). Fig. 6 visualises the contrast of the simulated versus observed egress times for each of the four scenarios while the parameter value (i.e. the social influence parameter) changes. The simulation outputs of each 100 simulation repetitions have been plotted in a box-plot format. It is clear from these comparisons that an avoid-the-crowd rule (i.e. negative coefficient value for the social influence parameter) far better describes and reproduces the observations compared to a follow-the-crowd modelling rule (i.e. positive coefficient value). In fact, as soon as the social influence valuation enters the positive domain (or as soon as imitative tendency is introduced to the (probabilistic) behaviour), the outcomes incur a substantial amount of bias. For example, by increasing the degree of imitation tendency we could potentially predict more than a 100 s duration time for a collective escape scenario that was observed to take only about 40 s.

4. Conclusions

Using econometric models of exit direction choice and a multi-layer evacuation simulation platform, we simulated escape behaviour of human crowds in confined spaces under varying degrees of imitative (herding) tendency. The parametric model allowed us to vary the tendency to imitate directional choices of the crowd on a continuum while measuring aggregate metrics related to evacuation efficiency. These measures included average of individual (simulated) evacuation times and total (simulated) evacuation time. In order to ensure that the findings are not an artefact of testing specific scenarios, we tested a broad variety of scenarios (20 scenarios in total). These numerical tests included scenarios in which the geometry of the space was symmetric as well as scenarios in which geometry was asymmetric, scenarios in which all exits were visible as well as scenarios in which a subset of exits were invisible, scenarios in which exit widths were equal as well as scenarios in which exit widths were unequal. Through changing the spatial distribution of exit widths, we indirectly created scenarios in which exit widths were highly limiting and thus the space became highly crowded as opposed to the scenarios in which the widths were less limiting and the crowdedness in the simulated environment was comparatively less heavy. The numerical tests showed that under none of those scenarios, a crowd of evacuees could benefit from amplified levels of herding tendency. According to our numerical tests, elevated herding tendencies in direction choice-making could potentially result

Table 2
Details of the limited experimental tests used for the macro-level analysis.

Scenario No.	Treatment type	No. of Exits	No. of subjects	Exit widths (cm)	Entrance width (cm)	Total egress time (sec)
1	Stressful	3	117	50	150	41
2	Non-stressful	3	117	50	150	48
3	Stressful	4	117	50	150	32
4	Non-stressful	4	117	50	150	41

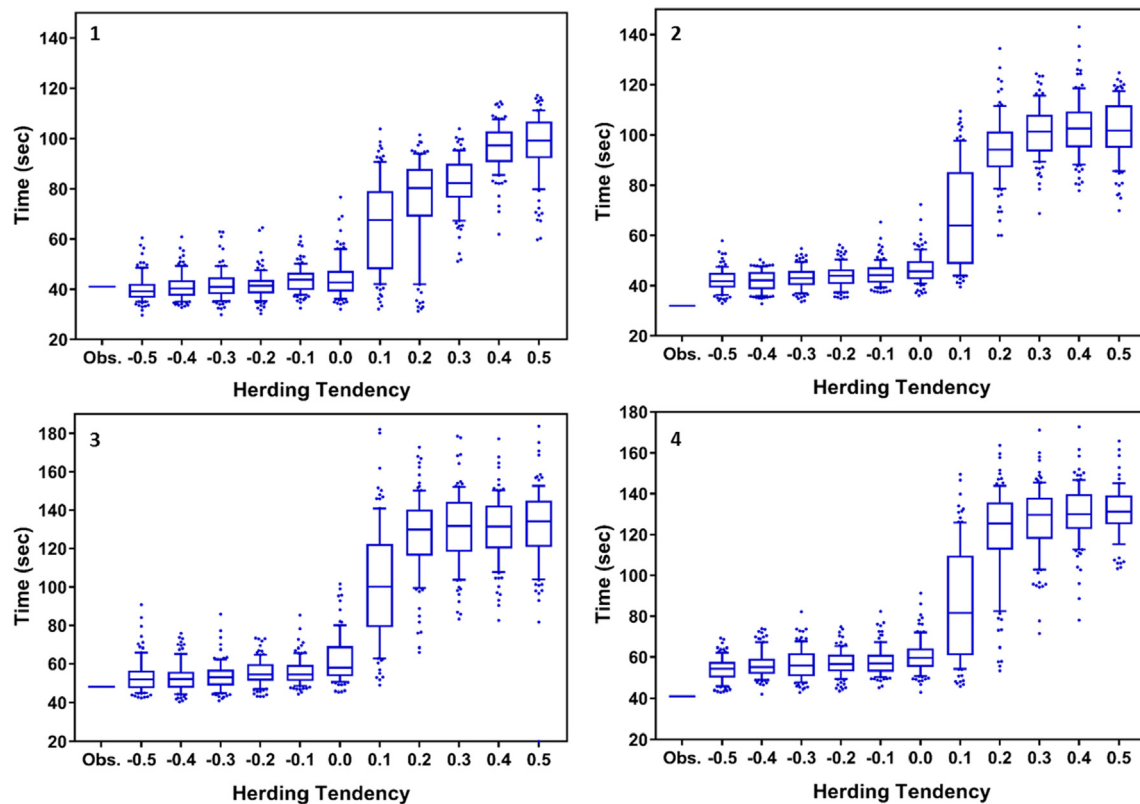


Fig. 6. Contrasting the observed total evacuation times (the horizontal lines) and the simulated evacuation times (the box plots) for each experimental scenario and under different levels of simulated herding tendency. Each simulation scenario has been repeated 100 times for each level of herding tendency.

in evacuation times of more than 100% bigger magnitude (compared to a scenario where the crowd shows a low degree of herding tendency or even an opposite tendency to herding). This suggested that in evacuation scenarios in crowded spaces, individuals should be dissuaded from adopting imitative strategies in making their escape direction choices. This, of course, is subject to the conditions in which occupants are least partially familiar with the surrounding environment and are at least aware of the location of exits. The findings may not be readily extendable to scenarios in which individuals are completely unfamiliar with the environment that they evacuate (Kinader et al., 2018; O'Neill, 1992).

As a secondary aspect of our analyses, we also utilised a limited number of experimental scenarios in which the social influence was the dominant factor in decision making. Using the aggregate measures (i.e. total evacuation time) obtained from these experimental scenarios, we tested various degrees of imitative tendency in direction choosing while replicating those scenarios in a computer-simulated setting. Results showed how mis-specifying the imitative tendency of the simulated agents could result in gross misestimation of the evacuation time. This provided further evidence pointing at the necessity of fine-tuning the tendency of evacuees to imitate (or not imitate) in order to produce realistic outcomes and highlighted the extent of modelling consequences of parameter mis-estimation on this aspect.

5. Discussion

This study was mainly motivated by suggestions (mostly speculative) that there may be scenarios and situations where a crowd of evacuees can benefit from herd-type behaviour (Pan et al., 2007). We also were motivated by studies that raised the question in a different way asking about the extent to which “peer imitation can affect efficiency of egress” (Moussaïd et al., 2016). The current study can be regarded as a direct follow-up to this question and a first attempt

towards addressing this aspect of evacuation optimisation. We believe that this question may have, to some degrees, been triggered and influenced by findings in economic literature that have demonstrated and discussed the possibility of “rational herding in financial markets” (Devenow and Welch, 1996). Therefore, we believe that it was fair to ask whether such rational herding can be observable in crowd evacuation contexts as well. We tested an extensive variety of scenarios (in terms of the geometry shape, crowding level etc) and failed to observe any particular situation where a crowd of evacuees (familiar with the position of exit locations) can benefit from such imitative tendencies in their direction choices. The findings pointed overwhelmingly to the opposite direction. The numerical results showed that such behaviour almost invariably hampers the efficiency of the system, often to substantial degrees.

We had argued in a previous study that the herding phenomenon and its effect on a system is not a universal phenomenon. It is rather highly context specific and depending on the context of the choice and the characteristics of the system, it can have different consequences (Haghani and Sarvi, 2017a). Therefore, it is of great importance to study this phenomenon in each context of decision making separately and remain cautious about overgeneralisation across the contexts as those may not be necessarily valid. People may be observed to show imitative tendencies in certain contexts of decision making (such as investment decisions), but that does not necessarily indicate they will do so in all contexts of choice. Their imitative tendency might be of some benefits (to individuals or the system) in certain contexts but the effect might be completely different and opposite for other systems. We believe that these findings were another testament to the context-specificity of this phenomenon.

Further to the above discussion, it should also be noted that evacuation and escape behaviour itself is not limited exclusively to one type of decision. An act of evacuation entails a range of decisions that need to be made by each individual (Haghani and Sarvi, 2018b;

Kuligowski, 2013, 2009; Kuligowski and Gwynne, 2010; Ronchi and Nilsson, 2013). In addition to exit direction decision making (Haghani and Sarvi, 2017d; Lovreglio et al., 2016a), an evacuation strategy choosing includes (but is not limited to) decision as to when initiate the evacuation movement (Bode and Codling, 2018; D'Orazio et al., 2015; Galea et al., 2017; Haghani et al., 2018b; Kuligowski and Mileti, 2009; Lovreglio et al., 2015), decision as to which global or local path to choose (Asano et al., 2010; Cristiani and Peri, 2017; D'Orazio et al., 2014), and decision as to whether to update/change initial directions choices (or any of other path-finding levels of choice) (Bode and Codling, 2013; Liao et al., 2017). In all those aspects of decision making, individuals are subject to the decision of their peers and may as well be affected by this social influence, while the direction and magnitude of this influence can vary depending on the nature and type of the choice. It is therefore important to recognise the role of decision type and clearly specify it when discussing the social influence in evacuations and to study this factor separately in relation to each aspect of the behaviour. In other words, the question of whether humans display imitative tendencies in emergency escape scenarios appears to be a rather oversimplified question which needs to be broken down into specific types and contexts and evacuation decision making.

Similarly, when raising the question of “optimum egress” (Teknomo and Fernandez, 2012) in connection to the role of social influence on system efficiency, it is also important to differentiate between various levels of decision making. Our findings in this study were exclusively related to the choice of exit direction and cannot be reasonably transferred to other aspects of evacuation decision making. Questions such as the followings are still open for exploration in this domain: Can peer imitation be beneficial in the decision of movement initiation? Can peer imitation be beneficial in terms of decision changing? Is it possible that observing other people (who are in the similar condition to the condition of the choice maker) changing their decisions makes the choice maker likelier to change his/her decision and follow them? In other words, do people show herding tendencies in their decision “changes”, and more importantly is there any benefit in adopting such strategy? We believe that numerical analyses using parametric models of behaviour (such as the case of this study) offer great potential for answering questions like this that can have real implications for evacuation management and training.

Another factor that needs to be clearly mentioned in relation to the simulation assumptions underlying the analyses of this study is the homogeneity of the crowd. Here, in this work we used probabilistic econometric models of choice for simulating direction choices of agents under varying levels of imitation tendency. Although different agents make different decisions as a result of the probabilistic nature of the underlying model, the choices are simulated based on an assumption of a behaviourally homogenous crowd with identical utility distributions (or, with identical distribution of escape strategy). In other words, such analysis disregards heterogeneity (or variation) in preferences (or strategies) or what is also known as ‘individual differences’ (Haghani and Sarvi, 2017b, 2018a) or ‘mixture of strategies’. The effect of individual differences in direction decision-making has been evidenced by earlier work suggesting that samples of evacuees are composed of multiple discrete classes in terms of how they evaluate (or prioritise) various attributes in their direction decision making. The case of heterogeneous (or multi-class) crowds (i.e. crowds with mixture of direction choice-making strategy) could be another dimension that can be explored when quantifying the effect of peer imitation on evacuation efficiency: Is there any benefit in having only a proportion of the escaping crowd follow the majority while the rest adopt an avoid-the-crowd strategy? In other words, could a crowd potentially benefit from a mixture of direction-choice strategy when it comes to the peer behaviour effect? And if so, what would the optimum composition be?

In this work, we also discussed and demonstrated the significance of fine-tuning the peer imitation tendency when simulating evacuation procedures. By linking between numerical analyses and experimental

observations we realised that extreme or even large magnitudes of imitation tendency produces simulated outputs that are very far from that of realistic observations. This is in line with the emerging body of empirical evidence that is also indicating that individuals do not tend to display noticeable or strong imitative tendencies in their direction choices (Bode and Codling, 2013; Haghani and Sarvi, 2017a; Li et al., 2016). It has also been further suggested that imitative tendencies are not universally constant and depending on the context (or individual)-specific factors may vary. Increased uncertainty in the environment for example has been shown to be one of those factors that moderate peer behaviour perception (Haghani and Sarvi, 2017a). Identifying these factors or their direction of impact on evacuation behaviour was outside the main scope of this study since, here, we only focused on the question of system efficiency while simulating ‘synthesised’ degrees of imitative tendency. However, in identifying the ‘natural’ level of imitative tendency that is itself a crucial aspect in terms of the accuracy of evacuation modelling, one should recognise the role of other possible moderating factors. Factors such as the ‘motivation’ (or ‘stress’) level, the level of general ‘crowding’ in the environment, or the degree of ‘familiarity’ with the environment can be thought of as potential moderating variables whose role need to be investigated by future work on this topic (Haghani and Sarvi, 2018c).

The social interactions have been proved to be only part of the mechanism of decision making. Humans make trade-offs between the physical and social factors and when the former outweighs the latter in their trade-off, they may ignore the social influence and place their decisions based on optimising the physical attributes (like choosing the nearest exit, if the benefit of that highly outperforms the social factor attributes). Therefore, similar question to that of our study can be raised as to the optimum level of valuation for physical factors. We observed that placing positive weights on the social factor is detrimental and a negative valuation of this factor (even at extreme levels) is actually more beneficial. One can raise the similar question about the valuation of the physical factors. For example, how about if in their strategy, humans place extreme weights on choosing the nearest exit? Would that be less optimal than a multi-attribute strategy? Or more importantly, what would be the general optimum behaviour (in terms of valuation of all influential attributes)? Answering such questions may ultimately translate in simple general advice or training for evacuations which, if followed by evacuees, benefits the system as a whole. This would constitute a first attempt in demystifying the notion of “irrational behaviour” (Zheng and Cheng, 2011) that is often used in this context as another speculative but common term in relation with humans’ escape behaviour. In answering that, we suggest that it would also be of crucial importance to be able to distinguish between the rationality (or optimality) from the system and individual-level perspectives (an aspect that would add even more layers of complications to this underexplored problem).

The authors wish to present their closing discussion on the use of terminology, ‘herding’, in describing what is meant to be an “imitative” type of behaviour. In the introduction section, we pointed out the prevalence of the use of this term in this context and others for referencing to ‘imitation’. Our use of this term was also in recognition of this precedent in the literature. However, we believe that the use of an absolute term like ‘herd’ (that as a noun is often used in reference to groups of animals and as a verb primarily refers to “moving in groups” (Mayor, 2009)) might not be perfectly accurate for describing a multifaceted aspect of behaviour for humans at least in the context of this work. As mentioned earlier in the introduction, the term does not convey imitative behaviour in its literal definition, but the scientific literature has largely adopted this term for this purpose and the term is now entrenched and has widespread and accepted use in the literature. But at least in the context of evacuation, we believe there is a specific downside to the use of this terminology. When we talk about ‘herd behaviour’ in evacuations it implicitly conveys the impression of a behaviour that is completely driven and dominated by imitation.

Whereas, recent empirical evidence in this field has shown time and time that decision-making of humans in escape scenarios is most likely not a single-attribute type of decision (Haghani et al., 2016; Kinatader et al., 2018). Evidence is increasingly suggesting that these decisions are best described by multi-attribute trade-offs between a range of factors of which social influence is only one item (Bode et al., 2014; Haghani and Sarvi, 2017a; Kinatader et al., 2014a, 2014b). People do consider the peer behaviour as a factor in their own decision (regardless of how they do so and whether they tend to follow or avoid those decisions, which is a different question) but at the same time, they also take into account other attributes such as minimising their distances or uncertainty of their choices to name a few. We suggest that the use of the term ‘herding behaviour’ as opposed to ‘imitation tendency’ is to large degrees dismissive of this multi-attribute trade-off aspect of the decision-making and may be an oversimplified term to describe a complicated phenomenon. Furthermore, it also dismisses the findings of recent studies indicating that the observation of peer behaviour does not necessarily trigger imitation. Rather, in many cases it triggers the opposite “avoid-the-crowd” type of behaviour (Haghani and Sarvi, 2017a). We believe that it is still a completely legitimate and useful question to investigate the degree of social influence or the direction of influence. But in terms of the choice of terminology, we also believe that this research field could potentially benefit from adopting an alternative terminology for describing this phenomenon, a term that firstly is neutral in terms of the direction of impact that it implies and secondly, a term that recognises the role of other factors besides peer behaviour that influence evacuation decision making.

Acknowledgements

The authors acknowledge that this study was financially supported by Discovery Project research grant DP160103291 awarded by Australian Research Council.

The authors extend their sincere gratitude to the editor and two anonymous reviewers of this article for their insightful remarks.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ssci.2018.12.026>.

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